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C*-algebraic Gauss-Lucas Theorem and C*-algebraic Sendov's Conjecture

K. M. Krishna*

School of Mathematics and Natural Sciences, Chanakya University Global Campus, NH-648, Haraluru Village, Devanahalli Taluk, Bengaluru North District, Karnataka State, 562 110, India

Abstract. Building on a foundational result established by Robertson [*Proc. Edinburgh Math. Soc., 1976*], we develop a framework for differentiation of maps defined on specific classes of unital commutative C*-algebras. This construction allows us to extend classical analytic ideas into the C*-algebraic setting, thereby enriching the interplay between operator theory and complex analysis. Within this framework, we derive a C*-algebraic analogue of the Gauss–Lucas theorem, which traditionally describes the geometric location of polynomial zeros in relation to their derivatives. Furthermore, we propose a C*-algebraic version of Sendov's conjecture, a long-standing problem in complex analysis concerning the proximity of critical points to polynomial zeros. Our formulation adapts this conjecture to the algebraic context and provides new insights into its structure. As a concrete verification, we establish the validity of the C*-algebraic Sendov's conjecture for all polynomials of degree two, thereby offering evidence for its broader applicability.

Key words: Sendov's conjecture, C-algebra, Gauss-Lucas theorem.*

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***Correspondence:** E-mail: kmaheshak@gmail.com

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Научная статья

Полный текст на английском языке

УДК 517.98

C*-алгебраическая теорема Гаусса-Лукаса и C*-алгебраическая гипотеза Сендова

К. М. Кришна*

Факультет математики и естественных наук Главный кампус Университета Чанакья, NH-648, деревня Харалуру, Северный округ Бангалора, Штат Карнатака, 562110, Индия

Аннотация. Опираясь на фундаментальный результат, установленный Робертсоном в 1976 г., мы разрабатываем структуру для дифференцирования отображений, определенных на конкретных классах унитарных коммутативных C*-алгебр. Эта конструкция позволяет нам расширить классические аналитические идеи на C*-алгебраическую среду, тем самым обогащая взаимодействие между теорией операторов и комплексным анализом. В рамках этой структуры мы выводим C*-алгебраический аналог теоремы Гаусса–Лукаса, которая традиционно описывает геометрическое расположение нулей многочленов относительно их производных. Кроме того, мы предлагаем C*-алгебраическую версию гипотезы Сендова, давней проблемы в комплексном анализе, касающейся близости критических точек к нулям многочленов. Наша формулировка адаптирует эту гипотезу к алгебраическому контексту и дает новые представления о ее структуре. В качестве конкретного подтверждения мы доказываем справедливость C*-алгебраической гипотезы Сендова для всех многочленов второй степени, тем самым предоставляя доказательства ее более широкой применимости.

Ключевые слова: гипотеза Сендова, C-алгебра, теорема Гаусса-Лукаса.*

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***Корреспонденция:** E-mail: kmaheshak@gmail.com

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Introduction

Let a be a complex number, $r > 0$ be a real number and $\overline{\mathbb{D}(a,r)}$ be the closed disc in $\mathbb{C}$ centered at a of radius r. Let $\mathbb{C}[z]$ be the set of all polynomials over $\mathbb{C}$. In 1958, Sendov made the following conjecture which became known as Sendov's conjecture.

Conjecture 1. *[1–3] (Sendov's conjecture) Let $n \in \mathbb{N} \setminus \{1\}$ and $p(z) = (z-a_1)(z-a_2)\cdots(z-a_n) \in \mathbb{C}[z]$ be such that $a_1, a_2, \ldots, a_n \in \overline{\mathbb{D}(0,1)}$. Then for each a_j, $1 \leq j \leq n$, there exists a zero b of p' such that $b \in \overline{\mathbb{D}(a_j,1)}$.*

A direct calculation says that Conjecture 1 holds for degree two polynomials. For $n = 3$, Conjecture 1 is proved in [4]. For $n = 3,4$, Conjecture 1 is proved in [5]. For $n = 5$, Conjecture 1 is proved in [6]. For $n = 6$, Conjecture 1 is proved in [7,8]. For $n = 7$, Conjecture 1 is proved in [9]. For $n = 8$, Conjecture 1 is proved in [10]. Conjecture 1 is proved for all polynomials after certain (unknown) degree in [11]. On the other hand, Conjecture 1 is proved for various special cases of polynomials and position of roots (inside the closed unit disc) in [5, 12–39]. Non-Archimedean version of Conjecture 1 has been solved in [40].

A subsection of noncommutative geometry strives to obtain complex analysis results by replacing $\mathbb{C}$ with a C*-algebra. A few prominent articles in this way are [41–44]. As C*-algebras offer greater challenge and Sendov conjecture is still open for vast set of polynomials, we believe that by generalizing the problem itself, we may get some light on six decades old conjecture itself.

In this paper, we derive Gauss-Lucas theorem for polynomials over certain C*-algebras. We then formulate Sendov's conjecture for polynomials over C*-algebras.

C*-algebraic Gauss-Lucas Theorem and C*-algebraic Sendov's Conjecture

Let $\mathcal{A}$ be a unital commutative C*-algebra and let $G(\mathcal{A})$ be the set of all invertible elements of $\mathcal{A}$. Let $\mathcal{A}[z]$ be the set of all polynomials over $\mathcal{A}$. We assume $G(\mathcal{A})$ is dense in $\mathcal{A}$. Such C*-algebras exist and a characterization of such C*-algebras is given by Robertson [45]. We now introduce the following definition.

Definition 1. *(C*-algebraic differentiation) Let $\mathcal{A}$ be a unital commutative C*-algebra and $G(\mathcal{A})$ be dense in $\mathcal{A}$. Let $f : \mathcal{A} \to \mathcal{A}$ be a function and $\omega \in \mathcal{A}$. We say that f is C*-algebraic differentiable at ω if there exists an $L \in \mathcal{A}$ satisfying the following: for each $\varepsilon > 0$, there exists a $\delta > 0$ such that if $z \in \mathcal{A}$ satisfies $\|z-\omega\| < \delta$ and $z - \omega \in G(\mathcal{A})$, then*

$$\|(z-\omega)^{-1}(f(z) - f(\omega)) - L\| < \varepsilon.$$

In this case, we write $f'(\omega) = L$.

Note that Definition 1 is nontrivial for any commutative C*-algebras which is not $\mathbb{C}$. The reason is unlike $\mathbb{C}$, nonzero elements need not be invertible. We note that, because $G(\mathcal{A})$ is dense in $\mathcal{A}$, Definition 1 is well-defined. Now we derive C*-algebraic Gauss-Lucas theorem.

Theorem 1. *(C*-algebraic Gauss-Lucas Theorem) Let* $p(z) = (z-a_1)(z-a_2)\cdots(z-a_n) \in \mathcal{A}[z]$. *Define*

$$\mathcal{B} \coloneqq \{z \in \mathcal{A} : z - a_j \text{ is invertible in } \mathcal{A} \text{ for all } 1 \le j \le n\}, \quad \mathcal{C} \coloneqq \{z \in \mathcal{A} : p'(z) = 0\}.$$

Then for each $z \in \mathcal{B} \cap \mathcal{C}$, *there are positive* $\omega_{z_1}, \ldots, \omega_{z_n} \in \mathcal{A}$ *such that*

$$z = \sum_{j=1}^{n} \omega_{z_j} a_j, \quad \sum_{j=1}^{n} \omega_{z_j} = 1. \tag{1}$$

Proof. We have

$$p'(z) = \sum_{j=1}^{n} (z-a_1)\cdots\widehat{(z-a_j)}\cdots(z-a_n), \quad \forall z \in \mathcal{A}.$$

where the term with cap is missing. Then

$$p(z)^{-1}p'(z) = \sum_{j=1}^{n} (z-a_j)^{-1}, \quad \forall z \in \mathcal{B}.$$

We therefore have

$$0 = \sum_{j=1}^{n} (z-a_j)^{-1} = \sum_{j=1}^{n} ((z-a_j)(z-a_j)^*)^{-1}(z^*-a_j^*), \quad \forall z \in \mathcal{B} \cap \mathcal{C}. \tag{2}$$

Now define

$$\omega_z \coloneqq \sum_{j=1}^{n} ((z-a_j)(z-a_j)^*)^{-1}, \quad \forall z \in \mathcal{B} \cap \mathcal{C}.$$

Note that whenever $z \in \mathcal{B}$, each $((z-a_j)(z-a_j)^*)^{-1}$ is positive and invertible, $1 \le j \le n$. Hence ω_z is positive and invertible. Equation (2) then gives

$$z = \omega_z^{-1} \sum_{j=1}^{n} ((z-a_j)(z-a_j)^*)^{-1} a_j, \quad \forall z \in \mathcal{B} \cap \mathcal{C}.$$

□

Definition 2. *Given a unital C*-algebra* $\mathcal{A}$ *with identity* 1 *and an element* $a \in \mathcal{A}$, *we define the C*-algebraic closed unit disc centered at* a *and of radius* $r > 0$, $r \in \mathbb{R}$, *denoted as* $\overline{\mathbb{D}^*(a,r)}$ *by*

$$\overline{\mathbb{D}^*(a,r)} \coloneqq \{z \in \mathcal{A} : (z-a)(z-a)^* \le \sqrt{r} \cdot 1\}.$$

With Definition 2 we can formulate Conjecture 1 for C*-algebras.

Conjecture 2. *(Commutative C*-algebraic Sendov's conjecture) Let* $\mathcal{A}$ *be a unital commutative C*-algebra and* $G(\mathcal{A})$ *be dense in* $\mathcal{A}$. *Let* $n \in \mathbb{N}\setminus\{1\}$ *and* $p(z) = (z-a_1)(z-a_2)\cdots(z-a_n) \in \mathcal{A}[z]$ *be such that* $a_1, a_2, \ldots, a_n \in \overline{\mathbb{D}^*(0,1)}$. *Assume that* p' *admits roots in* $\mathcal{A}$, *say* $b_1, b_2, \ldots, b_{n-1} \in \overline{\mathbb{D}^*(0,1)}$ *and each* b_k *can be written in*

the form of Equation (1). Then for each a_j, $1 \leq j \leq n$, *there exists a zero* b *of* p' *such that* $b \in \overline{\mathbb{D}^*(a_j, 1)}$.

For arbitrary C*-algebras, without introducing differentiation we can formulate Conjecture Conjecture 2 as follows.

Conjecture 3. *(C*-algebraic Sendov's conjecture) Let* $\mathcal{A}$ *be a unital C*-algebra. Let* $n \in \mathbb{N}\backslash\{1\}$ *and* $p(z) = (z-a_1)(z-a_2)\cdots(z-a_n) \in \mathcal{A}[z]$ *be such that* $a_1, a_2, \ldots, a_n \in \overline{\mathbb{D}^*(0,1)}$. *Define*

$$p'(z) = \sum_{j=1}^{n} (z-a_1)\cdots\widehat{(z-a_j)}\cdots(z-a_n), \quad \forall z \in \mathcal{A}.$$

where the term with cap is missing. Assume that p' *admits roots in* $\mathcal{A}$, *say* $b_1, b_2, \ldots, b_{n-1} \in \overline{\mathbb{D}^*(0,1)}$ *and each* b_k *can be written in the form of Equation (1). Then for each* a_j, $1 \leq j \leq n$, *there exists a zero* b *of* p' *such that* $b \in \overline{\mathbb{D}^*(a_j, 1)}$.

We can formulate the following variant of Conjecture 3.

Conjecture 4. *(C*-algebraic Sendov's conjecture - 2) Let* $\mathcal{A}$ *be a unital C*-algebra. Let* $n \in \mathbb{N} \setminus \{1\}$ *and* $p(z) = (z-a_1)(z-a_2)\cdots(z-a_n) \in \mathcal{A}[z]$ *be such that* $a_1, a_2, \ldots, a_n \in \overline{\mathbb{D}^*(0,1)}$. *Assume that* p' *completely factors into linear factors. Then for each* a_j, $1 \leq j \leq n$, *there exists a zero* b *of* p' *such that* $b \in \overline{\mathbb{D}^*(a_j, 1)}$.

Note that Conjectures 2, 3 and 4 reduce to Conjecture 1 whenever C*-algebra is $\mathbb{C}$. In fact, nonzero complex numbers are invertible and dense so Conjecture 2 reduces to Conjecture 1. It is obvious that for complex numbers Conjecture 3 reduces to Conjecture 1. Since every polynomials over $\mathbb{C}$ splits into linear factors, it is clear that Conjecture 4 reduce to Conjecture 1 for complex numbers. We end with the following result.

Theorem 2. *Let* $\mathcal{A}$ *be a unital C*-algebra. Conjecture 3 holds for polynomials of degree 2 of the form* $p(z) = (z-a)(z-b) \in \mathcal{A}[z]$, *where* $a, b \in \overline{\mathbb{D}^*(0,1)}$.

Proof. We have

$$p'(z) = 2z - (a+b), \quad \forall z \in \mathcal{A}.$$

Therefore the zero of p' is $\frac{a+b}{2}$ which is in the form of Equation (1). Now we have

$$2(aa^* + bb^*) \geq (a+b)(a+b)^*, \quad 2(aa^* + bb^*) \geq (a-b)(a-b)^*$$

which gives

$$\left(\frac{a+b}{2} - a\right)\left(\frac{a+b}{2} - a\right)^* \leq 1, \quad \left(\frac{a+b}{2} - b\right)\left(\frac{a+b}{2} - b\right)^* \leq 1.$$

Therefore

$$\frac{a+b}{2} \in \overline{\mathbb{D}^*(a,1)}, \quad \frac{a+b}{2} \in \overline{\mathbb{D}^*(b,1)}$$

which completes the proof. □

We believe that Conjectures 2, 3 and 4 can be verified whenever zeros are positive invertible elements. We verified the C*-algebraic Sendov conjecture only for degree 2 polynomials. Like classical case, for degree 3-8, we believe that it can be solved by computations, at least for special polynomials.

We suggest the approach of proving our conjecture for C*-algebras generated by finitely many elements using functional calculus.

Conclusion

The paper establishes a C*-algebraic framework for classical results in complex analysis by defining differentiation on unital commutative C*-algebras where invertibles are dense, leveraging Robertson's result. It proves the C*-algebraic Gauss-Lucas theorem, showing that critical points of polynomials over such algebras lie in positive convex combinations of the roots. Additionally, it formulates two versions of the C*-algebraic Sendov's conjecture—one requiring differentiability and one without—and verifies the latter for degree-2 polynomials.

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Information about the author



Krishna Mahesh ✉ – PhD (Phys. & Math.), Assistant Professor, School of Mathematics and Natural Sciences, Chanakya University Global Campus, Haraluru, India, ORCID 0000-0003-4872-8634.